

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of experience as a Software Developer and Data Engineer, specializing in building scalable microservices and distributed data pipelines. Proficient in Java, Python, Spark, SQL, AWS, and Azure, with expertise in CI/CD automation, large-scale data platforms, and privacy-preserving AI applications. Leverages Grpc, Kafka, and RabbitMQ for robust system design and asynchronous programming, focused on performance tuning and efficient multithreading.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Automated CI/CD pipelines using Jenkins and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments with robust unit testing.
- Modernized legacy systems by migrating to Log4j 2.x and improving monitoring, reducing issue resolution time by 20 percent.
- Led a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, utilizing Git and SVN for version control and Cucumber for behavior-driven development.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered Java and EJB-based enterprise applications and RESTful web services for automotive clients, demonstrating strong system design capabilities.
- Improved system performance by 90 percent through SQL query optimization and database tuning on legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, ensuring efficient release cycles.
- Mentored junior developers through code reviews, debugging sessions, and clean coding practices, fostering a collaborative environment.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance tuning.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity using NoSQL databases concepts.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and ensuring code quality.
- Supported release planning and sprint execution using JIRA within Agile development teams, focusing on asynchronous programming patterns for efficient task management.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real-time, utilizing Kafka for message queuing.
- Engineered an interactive Flask web application for data exploration with dynamic filtering and visualization, integrating Grpc for efficient service communication.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and orchestration with Kubernetes.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent through advanced performance tuning.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, ensuring scalable infrastructure.

Privacy-Preserving AI Email Client (Capstone Project) (T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, leveraging Hugging Face libraries.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, exploring Heap memory analysis for optimization.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, demonstrating expertise in analyzing AI model performance.
- Explored Wasm for potential client-side inference, ensuring efficient execution and minimal overhead for the application.

SKILLS

- Programming Languages: Proficient in Java and Python, with extensive SQL experience for complex data analysis and backend development. Explored Threads and Multithreading for concurrent processing.
- Frameworks & Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, and

Cucumber for robust application development and testing.

- Web Services & Messaging: Skilled in designing and implementing REST APIs, SOAP APIs, Grpc, Kafka, and RabbitMQ for scalable and efficient communication systems.
- Cloud Platforms & Tools: Experienced with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform, Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for infrastructure automation and Orchestration.
- Concepts & Methodologies: Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, Privacy-preserving LLM-based applications, Analyzing, Benchmarking, System Design, Async Programming, Heap memory management, and performance tuning.